

# Stochastic thermodynamics

Work, heat, and rare trajectories

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A trapped bead or a molecular coordinate fluctuates substantially from one trajectory to the next. Thermodynamics still applies, but its bookkeeping must distinguish trajectory-level random quantities from ensemble inequalities. An overdamped coordinate in one thermal bath is a useful starting point.

## 1 A first law along a noisy path

Let  $x$  move in a potential  $U(x, \lambda)$  controlled by a prescribed protocol  $\lambda(t)$ , with an additional applied force  $f(x, t)$ . For constant friction  $\gamma$  and bath temperature  $T$ ,

$$\gamma \dot{x} = -\partial_x U + f + \sqrt{2\gamma k_B T} \xi(t), \quad \langle \xi(t) \xi(t') \rangle = \delta(t - t'). \quad (1)$$

The noise has zero mean. This model assumes separation from the bath's fast time scales and the Einstein relation for thermal noise; an active fluctuating force cannot simply be assigned the same temperature.

Define work done *on* the system and heat delivered *to* the bath by

$$\delta W = \partial_\lambda U d\lambda + f \circ dx, \quad (2)$$

$$\delta Q_{\text{bath}} = (-\partial_x U + f) \circ dx, \quad (3)$$

$$dU = \delta W - \delta Q_{\text{bath}}. \quad (4)$$

The circle denotes a Stratonovich integral. It permits the ordinary chain rule  $dU = \partial_x U \circ dx + \partial_\lambda U d\lambda$ . Even though the Langevin noise is additive, Itô and Stratonovich conventions differ for the heat integral because its integrand depends on  $x$ .

## 2 Entropy includes the probability of the state

Let  $p(x, t)$  be the ensemble probability density. Up to a fixed reference measure, the stochastic system entropy is  $s(x, t) = -k_B \ln p(x, t)$ . For a trajectory of duration  $\tau$ ,

$$\Delta s_{\text{tot}} = -k_B \ln \frac{p(x_\tau, \tau)}{p(x_0, 0)} + \frac{Q_{\text{bath}}}{T}. \quad (5)$$

The first term is essential when the distribution changes. Heat divided by temperature alone is not the total entropy production.

Under local detailed balance, the logarithm of the ratio between the forward path probability and the appropriately reversed path probability is  $\Delta s_{\text{tot}}/k_B$ . Here the reversed experiment uses the reversed protocol and starts in the forward final distribution; position is even under time reversal. Normalization of the reversed path measure gives the integral fluctuation theorem [1].

### 3 The second law from a fluctuation identity

For the regular diffusion above, with the required path-measure support,

$$\langle e^{-\Delta s_{\text{tot}}/k_B} \rangle = 1, \quad \langle \Delta s_{\text{tot}} \rangle \geq 0. \quad (6)$$

The inequality follows directly from Jensen's inequality. A single trajectory can have negative entropy production without contradicting it. The identity also implies, for any  $a > 0$ ,

$$\Pr(\Delta s_{\text{tot}} \leq -a) \leq e^{-a/k_B}, \quad (7)$$

by applying Markov's inequality to the nonnegative variable  $e^{-\Delta s_{\text{tot}}/k_B}$ . Large negative fluctuations are constrained quantitatively, not forbidden.

For purely potential driving,  $f = 0$ , prepare the initial state in canonical equilibrium and change  $\lambda$  at any finite rate. The inclusive work obeys the Jarzynski equality [2]

$$\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}, \quad \beta = (k_B T)^{-1}, \quad \langle W \rangle \geq \Delta F. \quad (8)$$

Here  $\Delta F$  compares the equilibrium free energies of the two endpoint potentials. The actual final distribution need not be equilibrated. This work relation and the total-entropy identity use different reversed initial ensembles; their exponents are not generally identical for a nonequilibrium final state.

### 4 An exactly checkable trap shift

Take  $U(x, \lambda) = \kappa(x - \lambda)^2/2$ , with  $\kappa > 0$ . Initially  $\lambda = 0$  and  $x_0$  is Gaussian with variance  $k_B T/\kappa$ . Shift the trap center instantaneously to  $a$ . Since the coordinate has no time to move,

$$W = \frac{\kappa}{2}(x_0 - a)^2 - \frac{\kappa x_0^2}{2} = \frac{\kappa a^2}{2} - \kappa a x_0. \quad (9)$$

Thus  $\langle W \rangle = \kappa a^2/2$  and  $\text{Var}(W) = \kappa a^2 k_B T$ . The Gaussian moment-generating function gives

$$\langle e^{-\beta W} \rangle = \exp \left[ -\beta \langle W \rangle + \frac{\beta^2}{2} \text{Var}(W) \right] = 1. \quad (10)$$

This agrees with  $\Delta F = 0$ : translating a harmonic trap leaves its partition function unchanged. There is no heat transfer during the instantaneous shift; dissipation occurs during subsequent relaxation. Rare negative-work realizations are needed for the exponential average to equal one.

Single-molecule pulling experiments motivate this distinction between work and free energy. In practice, exponential averages can converge slowly because rare trajectories carry large weight. Hidden coordinates, feedback, and chemical reservoirs require their own thermodynamic accounting before applying these formulas.

## References

- [1] U. Seifert, "Stochastic thermodynamics, fluctuation theorems, and molecular machines," *Rep. Prog. Phys.* **75**, 126001 (2012). [arXiv:1205.4176](#).
- [2] C. Jarzynski, "Nonequilibrium Equality for Free Energy Differences," *Phys. Rev. Lett.* **78**, 2690 (1997). [arXiv:cond-mat/9610209](#).